EQUAZIONI ALLE DERIVATE PARZIALI

ELLITTICO + PARABOLICO

PARAMETRICHE

$u : [a,b] \to \mathbb{R}$ $\qquad u \in \mathcal{C}^2([a,b])$

$$-u''(x) = f(x)$$

$\underline{f \equiv 1}$ $\qquad u = -x^2 + ax + b$

$u(0) = 0$

$u(1) = 0$

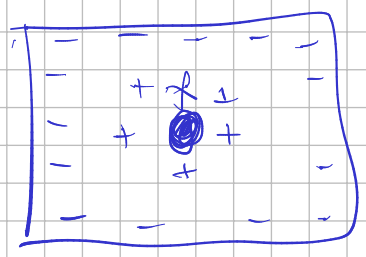
$$u(x) = \frac{1}{2}(1-x)x$$

$\Omega \subseteq \mathbb{R}^d \qquad d > 1 \qquad u : \Omega \to \mathbb{R}$

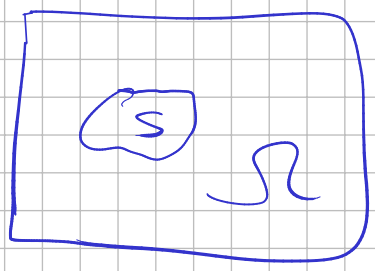
$-\Delta u(x) = f(x) \qquad \forall x \in \Omega$ $\qquad$ EQUAZIONE DI POISSON

$\Delta u := \partial^2_{x_1} u + \partial^2_{x_2} u + \dots \partial^2_{x_d} u$

$f \equiv 0 \Rightarrow$ "OMOGENEA" $\Rightarrow$ EQUAZIONE DI LAPLACE



$S \subset \Omega$



$$\underbrace{\int_{\partial S} F \cdot n \, d\Gamma = 0}$$

$$\parallel$$

$$\int_S \operatorname{div}(F)\, dx = 0 \qquad \forall\, S \subset \Omega$$

$$\operatorname{div}(F) = 0 \qquad F = -a \nabla u$$

$$\operatorname{div}(-a \nabla u) = 0 \qquad \forall\, x \in \Omega$$

$$-a \operatorname{div}(\nabla u) = 0 \qquad \operatorname{div}(F) = \partial_x F^1 + \partial_y F^2$$

$$\underline{-a \Delta u = 0} \qquad \operatorname{div}\begin{pmatrix} \partial_x u \\ \partial_y u \end{pmatrix} = \partial_x \partial_x u + \partial_y \partial_y u$$

$\Downarrow$

f (FIATTA)

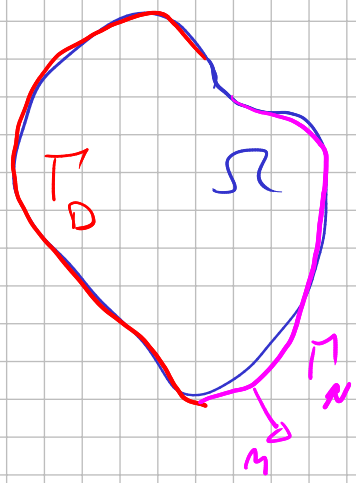
$\to$ SORGENTE

BORDO

- CONDIZIONI DI DIRICHLET

$$u(x) = g(x) \qquad \forall\, x \in \partial\Omega$$

$$x \in \underline{\underline{\Gamma_D}} \subset \partial\Omega$$



- CONDIZIONI DI NEUMANN

$$\frac{\partial u}{\partial \vec{n}} = \nabla u \cdot \vec{n} = h(x) \qquad \text{su} \quad x \in \Gamma_N \subset \partial\Omega$$

$x, t \qquad x \in \Omega \subseteq \mathbb{R}^d \qquad t \in \mathbb{R}^+$

$u : \Omega \times \mathbb{R}^+ \to \mathbb{R}$

$$\underline{\partial_t u(t,x)} - a \Delta u(t,x) = f(t,x,u)$$

IN 1D SPAZIALE + 1 TEMPO

$$\boxed{\partial_t u - a\, u_{xx} = f}$$

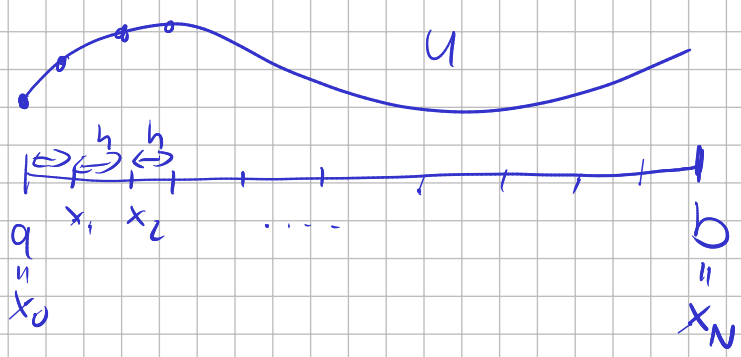
PROBLEMA DI CAUCHY

$\begin{cases} \text{PDE} \\ \text{BC} \\ \text{IC} \end{cases}$ BOUNDARY CONDITIONS, INITIAL CONDITIONS

$$\begin{cases} \partial_t u(t,x) - a\, \partial_{xx} u(t,x) = f(t,x,u) & \forall x \in \Omega \quad t \in \mathbb{R}^+ \\ u(0,x) = u_0(x) & \forall x \in \Omega \\ u(t,x) = u_D(t,x) & \forall t \in \mathbb{R}^+ \ \forall x \in \Gamma_D \subset \partial\Omega \\ \dfrac{\partial u(t,x)}{\partial n} = u_N(t,x) & \forall t \in \mathbb{R}^+ \ \forall x \in \Gamma_N \subset \partial\Omega \end{cases}$$

$$\boxed{\partial_t u + Au = F(u)}$$

DIFFERENZE FINITE



$u(x_0) \simeq u_0$

$u(x_1) \approx u_1$

$\vdots$

$u(x_N) \approx u_N$

$u_i \approx u(x_i)$

$x_{i+1} - x_i = h \quad \forall i = 0, \dots, N-1$

u' APPROSSIMATE

h

DIFFERENZE DIVISE

- IN AVANTI $\quad \delta_{h,+} u(x) = \dfrac{u(x+h) - u(x)}{h} \approx \partial_x u(x)$

- INDIETRO $\quad \delta_{h,-} u(x) := \dfrac{u(x) - u(x-h)}{h} \approx \partial_x u(x)$

- CENTRATA $\quad \delta_h u(x) = \dfrac{u(x+h/2) - u(x-h/2)}{h} \approx \partial_x u$

DERIVATA SECONDA ?

$$\delta_h^2 u(x) := \delta_{h,-}(\delta_{h,+} u)(x) = \delta_{h,-}\left(\frac{u(x+h) - u(x)}{h}\right)$$

$$= \frac{u(x+h) - u(x)}{h^2} - \frac{u(x) - u(x-h)}{h^2}$$

$$= \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$\delta_h^2 u - u''(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} - u''(x) =$$

$$= \frac{\cancel{u(x)} + \cancel{h u'(x)} + \frac{h^2}{2} u''(x) + \cancel{\frac{h^3}{6} u'''(x)} + \frac{h^4}{24} u^{(4)} - \cancel{2u(x)}}{h^2}$$

$$+ \frac{\cancel{u(x)} - \cancel{h u'(x)} + \frac{h^2}{2} u''(x) - \cancel{\frac{h^3}{6} u'''} + \frac{h^4}{24} u^{(4)}}{h^2} - u''(x)$$

$$= \underbrace{\frac{1}{2}u''(x) + \frac{1}{2}u''(x) - u''(x)}_{=0} + \frac{h^2}{24}u^{(4)} + \frac{h^2}{24}u^{(4)}$$

$$= \frac{h^2}{12}u^{(4)}(x) + O(h^3)$$

$$\left|\delta_h^2 u - u''\right| \le \frac{h^2}{12}\max_{\xi\in[a,b]}\left|u^{(4)}(\xi)\right|$$

$$-u''(x) = f$$

$$-u''(x_i) \approx -\frac{u(x_i-h) - 2u(x_i) + u(x_i+h)}{h^2}$$

$$\approx -\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2}$$

$$\forall x_i \rightsquigarrow \boxed{-\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = f(x_i) \qquad \forall i = 1, \dots, N-1}$$

$$-\frac{1}{h^2}\underbrace{\begin{array}{c} \\ \\ \\ \\ \\ \end{array}\begin{array}{cc} \begin{bmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & -2 \end{bmatrix} & \begin{array}{l} \leftarrow 1 \\ \leftarrow 2 \\ \\ \\ \leftarrow N-1 \end{array} \\ \begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ 1 & 2 & N-1 \end{array} & \end{array}}_{A} \underbrace{\begin{bmatrix} u_1 \\ \\ \\ u_{N-1} \end{bmatrix}}_{\vec{u}} = \underbrace{\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_{N-1}) \end{bmatrix}}_{F}$$

$$A\vec{u} = F$$

$\Rightarrow \boxed{u_0 = \alpha}$ $\boxed{\text{BORDI}}$ DIRICHLET $u_N = \beta$

$$-\frac{1}{h^2}\begin{bmatrix} -h^2 & 0 & & & 0 \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ 0 & & & 0 & -h^2 \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_N \end{bmatrix} = \begin{bmatrix} \alpha \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{N-1}) \\ \beta \end{bmatrix}$$

$$\| $$

$$\begin{bmatrix} 1 & 0 & & & 0 \\ -\frac{1}{h^2} & \frac{2}{h^2} & -\frac{1}{h^2} & & \\ & \ddots & \ddots & \ddots & \\ 0 & & -\frac{1}{h^2} & \frac{2}{h^2} & -\frac{1}{h^2} \\ & & & 0 & 1 \end{bmatrix}$$

$$\partial_x u(0) = \alpha$$

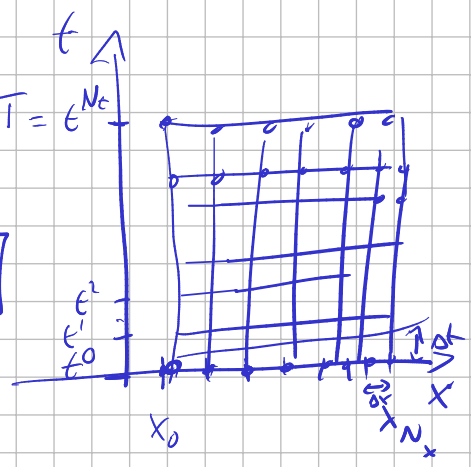
$$\frac{u_1 - u_0}{h} = \alpha \Rightarrow \begin{bmatrix} -\frac{1}{h} & \frac{1}{h} & 0 & & 0 \\ & \ddots & \ddots & \ddots & \\ \vdots & & & & \vdots \end{bmatrix} = \begin{bmatrix} \alpha \\ f(x_1) \\ \vdots \end{bmatrix}$$

$\boxed{\text{CALORE}}$

$$\partial_t u - \underbrace{\partial_{xx} u} = 0$$

$\Omega = [a,b]$ $\mathbb{R}^+$ $[0,T]$

$$u(x_i, t^n) \approx u_i^n$$



• EULERO ESPLICITO

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = 0$$

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x^2}\left(u_{i+1}^n - 2u_i^n + u_i^n\right)$$

• EULERO IMPLICITO

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} = 0$$

$$LHS = I - \frac{\Delta t}{\Delta x^2} D_2 = \begin{bmatrix} 1+2\frac{\Delta t}{\Delta x^2} & -\frac{\Delta t}{\Delta x^2} & & 0 \\ -\frac{\Delta t}{\Delta x^2} & \ddots & \ddots & \\ & \ddots & \ddots & \\ 0 & & & \end{bmatrix}$$

$$LHS \cdot \underline{u}^{n+1} = \underline{u}^n \qquad \forall\, n = 1, \dots, N_T$$

NONLINEARE

$$\underbrace{\partial_t u - \partial_{xx} u}_{} = \underbrace{u - u^3}_{}$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} = u_i^{n+1} - \left(u_i^{n+1}\right)^3$$

• COME RISOLVERE SISTEMA NON LINEARE?

METODO DI NEWTON!

$$R(\underline{u}^{n+1}) = 0 \qquad R : \mathbb{R}^{N_x} \to \mathbb{R}^{N_x}$$

$$JR \in \mathbb{R}^{N_x \times N_x}$$

$$(JR)_{ij} = \frac{\partial R_i(u^{n+1})}{\partial u_j^{n+1}} = \delta_{ij} + \Delta t \left(- \frac{\delta_{i+1,j} - 2\delta_{i,j} + \delta_{i-1,j}}{\Delta x^2} \right.$$

$$\left. - \delta_{ij} + \delta_{ij}\, 3 (u_i^{n+1})^2 \right)$$

$\forall n$ TEMPO

$\forall k$ ITERAZ NEWTON

$$\begin{cases} J_{u^{n+1}} R(u^{(k)}) \cdot \delta u^{(k)} = - R(u^{(k)}) \\ u^{(k+1)} = u^{(k)} + \delta u^{(k)} \end{cases}$$